A Nonasymptotic Good-Collar Bound on SU(2)

Finite-time prefactors and the one-link ultraviolet threshold

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Abstract

For the one-link heat-kernel conditional law with m coherent staples, we derive an explicit finite-time bad-curvature bound. The estimate combines radial monotonicity of the central SU(2) heat kernel, exact Haar ball volume, and a controlled Poisson-image ratio. It upgrades a small-time rate theorem by retaining every prefactor. At $\rho = 0.0082$, $\eta = 0.001$, and $m = 6$, the bound crosses the degree-18 connected-set weight at $t \simeq 8.9942 \times 10^{-7}$; at $t = 3 \times 10^{-6}$ its prefactor makes it trivial. The result assumes a coherent staple collar and does not estimate the probability of collar failure.

1 Heat-kernel input

Identify SU(2) with the unit three-sphere and write $\theta = d(I, g) \in [0, \pi]$. The heat kernel for generator $\frac{1}{2}\Delta$ is central, so it is denoted $p_t(\theta)$.

Lemma 1.1 (Radial monotonicity)**.** *For every $t > 0$, $p_t(\theta)$ is nonincreasing on $[0, \pi]$.*

Proof. For smooth radially decreasing approximations to the point mass, the radial heat equation on S^3 is

$$\partial_t u = \tfrac{1}{2}(u_{\theta\theta} + 2\cot\theta\, u_\theta).$$

Differentiate on compact subintervals of $(0, \pi)$ and apply the parabolic maximum principle to the positive part of u_θ, using the symmetry boundary values at 0 and π. Radial decrease is preserved. Passing to the heat-kernel limit gives the assertion. □

For $0 \le \theta \le R < \pi/2$, the paired Poisson-image representation gives

$$p_t(\theta) = C_t \,\frac{\theta}{\sin\theta} e^{-\theta^2/(2t)}(1 + r_t(\theta)), \qquad |r_t(\theta)| \le q_t(R) < 1. \tag{1}$$

The executable certificate bounds $q_t(R)$ by a convergent image tail. Put

$$c_t(R) = \frac{1 + q_t(R)}{1 - q_t(R)}.$$

Equation (1) implies

$$\frac{p_t(\rho)}{p_t(0)} \le \frac{\rho}{\sin\rho} c_t(R) e^{-\rho^2/(2t)}, \tag{2}$$

$$\frac{p_t(0)}{p_t(r_0)} \le c_t(R) e^{r_0^2/(2t)}. \tag{3}$$

2 Finite-time theorem

Let staples S_i lie in a common η-ball centered at C. After setting $V = UC$ and $Q_i = C^{-1}S_i$, the conditional law is proportional to $\prod_{i=1}^{m} p_t(VQ_i)\,\mathrm{d}V$, with $d(I, Q_i) \le \eta$. For target j let $B_j(\rho) = \{V : d(I, VQ_j) \ge \rho\}$.

The normalized Haar volume of the ball $d(I, V) \le \delta$ is

$$v(\delta) = \frac{\delta - \frac{1}{2}\sin(2\delta)}{\pi}. \tag{4}$$

Theorem 2.1 (Nonasymptotic coherent-collar bound)**.** *Assume*

$$R = \max\{\rho, \eta + \delta\} < \pi/2, \qquad \rho^2 > m(\eta + \delta)^2.$$

Then, uniformly over all coherent staple families,

$$\gamma_{t,S}(B_j(\rho)) \le \frac{1}{v(\delta)} \frac{\rho}{\sin\rho} c_t(R)^{m+1} \exp\left[-\frac{\rho^2 - m(\eta+\delta)^2}{2t}\right]. \tag{5}$$

The right side may always be replaced by its minimum with one.

Proof. On $B_j(\rho)$, radial monotonicity bounds the target factor by $p_t(\rho)$ and every other factor by $p_t(0)$. Thus the numerator is at most $p_t(\rho)p_t(0)^{m-1}$ because normalized Haar volume is one. In the ball $d(I, V) \le \delta$, the triangle inequality gives $d(I, VQ_i) \le \eta + \delta$, so the normalization is at least $v(\delta)p_t(\eta + \delta)^m$. Divide, insert (2)–(3), and simplify. □

Remark 2.2. *The auxiliary δ is a proof parameter. The reported golden-section search returns an admissible value and makes no theorem of global numerical optimality. Every returned probability remains rigorous for its selected δ.*

3 Calibration and decision

For $m = 6$, $\rho = 0.0082$, $\eta = 0.001$, $\alpha = 0.1$, and $\Delta = 18$, the critical connected-set weight is

$$q_{\mathrm{crit}} = e^{-\alpha}\Delta^{-2} = 0.00279270808036.$$

quantity	value
frozen time	3.0×10^{-6}
selected frozen δ	$8.22875586 \times 10^{-4}$
frozen raw log bound	14.97453169
finite-time probability upper bound	1
crossing time for q_{crit}	$8.99420304 \times 10^{-7}$
crossing δ	$3.36486781 \times 10^{-4}$

The favorable exponential rate at the frozen witness is insufficient after the denominator-volume prefactor is restored. The theorem nevertheless gives a concrete smaller-time regime where the one-link conditional estimate meets the desired counting weight.

4 Frontier boundary

The theorem is finite-volume, nonasymptotic, noncommutative, and gauge covariant through the common-ball condition. It does not show that coherent collars occur under arbitrary adaptive histories. It also does not control multi-link integration, repeated RG, continuum limits, or a Hamiltonian gap. The result is therefore a C2 restricted theorem.